MIL-C-24643/34B
3 October 1986
SUPERSEDING
MIL-C-0024643/34A(SH)
8 November 1985
MIL-C-24643/34
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSMS (INCLUDING VARIATION LSMSA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSMS (unarmored) and LSMSA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First	- Copper conductor, AWG no. 16, class B stranding, (7 by 0.0192 inch) or in accordance with ASTM B 286, size 16-19, tin coated. Maximum diameter 0.063 inch.
Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	- Ethylene propylene rubber or cross-linked polyethylene insulation, white. Minimum thickness 0.011 inch. Standard identification code applied by method 1.
Fourth	- The required number of singles (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward.
Fifth	- Binder tape, polyester, type G in accordance with MIL-I-631, applied helically with overlap.
Sixth	- Braided shield of AWG no. 34 or no. 36 tin coated copper. Minimum coverage of 85 percent.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSMS

Seventh - Cross-linked polyolefin jacket. Cable jacket thickness 0.060-inch minimum. Cable surface marking required.

Construction variant for type LSMSA

Seventh - Cross-linked polyolefin jacket. Cable jacket thickness 0.060-inch minimum.
Eighth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/34	Size 1/	Number of conductors	Overall diameter (inches)			
			Type LSMS		Type LSMSA	
			minimum	maximum	minimum	maximum
-01UO	37	37	0.740	0.800	0.790	0.850

1/ Unarmored, overall shield. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSMS and LSMSA.

NOTE: Weight of completed cable shall not exceed 0.60 pound per foot.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	5.1
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Insulation resistance - megohms/1000 feet minimum.....	500
Jacket flaws.....	No failure
Conductor and shield continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Cold bending, cable - at minus 30 ± 2°C mandrel diameter 10 inches, maximum.....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.....	250

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EXAMINATION AND TESTS: (Continued)

	Requirements
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum.....	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum.....	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing (conductor) - cycles, minimum....	50
Permanence of printing (jacket) (type LSMS only) - cycles, minimum.....	250
Heat distortion - percent of unaged, maximum.....	30
Cable filler removability.....	No failure
Armor (type LSMSA only) - conformance to material, construction and coverage.....	No failure
Shield conformance to material, construction and coverage.....	No failure
Surface transfer impedance	
Milliohms per meter, maximum.....	700
EMP response time - ms/500 feet.....	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C).....	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum.....	75

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-34)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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